BECM 3115

CLIMATE & ARCHITECTURAL DESIGN

SECTION B - MASTER CLASS NOTE

Exam-oriented | বাংলা ব্যাখ্যা | Memory shortcuts | Clean sketches | Matching final questions

Coverage

Introduction • Global Climatic Factors • Elements of Climate • Driving Rain / Solar / Wind • Tropical Climate Classification • Regional & Site Climate • Thermal Comfort

Prepared from uploaded Section B lecture slides + Billah Sir class notes + Regular Final Question Bank (2016-2024) + uploaded reference book for cross-checking.

Made by 2223033

এই নোট কীভাবে ব্যবহার করবে | How to Use This Note

Exam Strategy

প্রতিটি topic-এ আগে "Core idea" বুঝবে, তারপর table / sketch দেখে answer structure মনে রাখবে। Question Bank Match দেখে কোন wording বেশি এসেছে বুঝে 10/15/20 marks অনুযায়ী depth বাড়াবে।

Answer-writing pattern

1. Definition / core concept (1-2 lines)
2. Mechanism / factors / classification (table বা numbered points)
3. Necessary sketch / graph / flow diagram
4. Building-design significance / application
5. One-line conclusion linking climate design comfort /energy.

Source discipline

- Section B lecture slides ও Billah Sir class note হলো primary source.
- Question Bank (2016-2024) ব্যবহার করা হয়েছে matching-question mapping-এর জন্য.
- Reference book কেবল lecture slide সংক্ষিপ্ত হলে clarification/cross-check-এর জন্য ব্যবহার করা হয়েছে.
- Course-specific numerical values (যেমন solar constant 1395 W/m²) lecture slide অনুযায়ী রাখা হয়েছে—exam consistency-এর জন্য।

Lecture / Topic Map

Lecture	Core topics	Exam character
01	Climatic study, building variation, bioclimatic architecture	Short theory + concept
02	Global climate, solar radiation, Earth-Sun relation, thermal balance, winds, ITCZ	Very high priority; diagrams
03	Temperature, humidity, vapour pressure, precipitation, climatic data	Measurement + definitions
03.1	Driving rain, sky condition, solar duration/intensity, wind, wind rose	Short note + formula + graph
04	Tropical climate classification: warm-humid, hot-dry, composite	Comparison + characteristics
05	Regional/site climate, factors, local deviation, urban climate	Repeated theory
06	Thermal comfort, metabolism, heat gain/loss, balance, variables	Repeated theory + equation

LECTURE 01 — Climatic Study & Bioclimatic Architecture

1. Climatic Design কেন দরকার?

Climatic design হলো এমন design approach যেখানে building-কে স্থানীয় climate-এর সাথে “fit” করানো হয়, যাতে comfort পাওয়া যায় এবং mechanical energy demand কমে। Lecture slide-এর মূল কথা: “Building must be designed to fit in with their climate.”

Climatic study কী সাহায্য করে	Design implication
Comfortable indoor climate	temperature, airflow, solar exposure নিয়ন্ত্রণ
Energy-saving structural design	passive control আগে; mechanical system পরে
Protection from external factors	sun, wind, rain, humidity অনুযায়ী envelope
Natural resource utilisation	sunlight, breeze, vegetation, site features ব্যবহার
Regional architectural character	ভিন্ন climate → ভিন্ন form, opening, material, method

Shortcut / মনে রাখার কৌশল — C-E-P-N

Comfort → Energy saving → Protection → Natural resources. Climatic study-এর benefits এই ৪ শব্দে মনে রাখো।

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2016	10	Importance of climate study for a BECM graduate.
2017	15	Where can a BECM professional use learning from Climate & Architectural Design?

2. কেন building form এক জায়গা থেকে আরেক জায়গায় বদলায়?

- Climate condition building arrangement/design-কে প্রভাবিত করে।
- Requirement determination (কী পরিমাণ shade/ventilation/thermal protection দরকার) climate নির্ভর।
- Equipment selection climate-load-এর সাথে সম্পর্কিত।
- Building method, material এবং formation regional climate-এর প্রতিক্রিয়া।

Core idea

Same function ≠ same building. Climate বদলালে form, envelope, opening, roof, material এবং orientation বদলাতে পারে।

3. Bioclimatic Architecture

Bioclimatic বলতে lecture slide-এ “relationship between living organisms and architecture” বোঝানো হয়েছে। অর্থাৎ nature/environment-এর logic ব্যবহার করে building form ও performance optimize করা।

- Nature-inspired response, কিন্তু শুধুই aesthetic imitation নয়।
- Sun, wind, terrain, vegetation ও seasonal conditionকে design input হিসেবে নেয়।
- Goal: comfort + lower energy demand + better environmental fit.

Source-note

Introduction slide ecological consideration-কে syllabus-এর অংশ হিসেবে দেখায়; তবে এই slide set-এ ecological consideration-এর আলাদা detailed lecture নেই। তাই এখানে unsupported detail যোগ করা হয়নি।

LECTURE 02 — Global Climatic Factors

1. Climate vs Weather

Term	Meaning	Time scale
Climate	একটি স্থানের দীর্ঘ সময়ের weather pattern	প্রায় 20-30 years
Weather	নির্দিষ্ট সময় ও স্থানের atmospheric condition	day-to-day / short term

Shortcut / মনে রাখার কৌশল — Weather = Today, Climate = Trend

আজ কী হচ্ছে = weather; দীর্ঘমেয়াদে সাধারণত কী হয় = climate।

2. Latitude, Altitude & Land-Water Relationship

- Latitude = Equator-এর উত্তর/দক্ষিণে angular distance. Low latitude বেশি solar exposure পায়; high latitude তুলনামূলক ঠান্ডা।
- Altitude = mean sea level-এর ওপরে উচ্চতা. Lecture slide: air rises becomes cooler; তাই পাহাড়ের ওপর নিচের তুলনায় ঠান্ডা।
- Large water body climate moderate করে, কারণ land-এর তুলনায় water ধীরে heat up/cool down করে।
- Tropical environment roughly 23.5°N থেকে 23.5°S; lecture slide-এ rainforest, monsoon, savanna climate উল্লেখ করা হয়েছে।

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2018	15	Why is it always colder at higher altitudes? Discuss with examples.
2024	12	Why is there a large day-night temperature difference in hot desert / subtropical regions?

3. Eleven Global Climatic Factors — master list

No.	Factor
1	Solar radiation quality
2	Solar radiation quantity
3	Tilt of the Earth's axis
4	Radiation at the Earth's surface
5	Earth's thermal balance
6	Winds: thermal forces
7	Trade winds: Coriolis force
8	Mid-latitude westerlies
9	Polar wind
10	Annual wind shifts
11	Influence of topography

Shortcut / মনে রাখার কৌশল — 1-5 = Sun/Earth, 6-10 = Wind, 11 = Topography

Global factorsকে তিন ব্লকে ভাঙো: (1-5) solar/earth processes, (6-10) circulation/winds, (11) topographic modification.

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

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Year	Marks	Matching question / focus
2016	25	Analyze major factors of global climate variations in different geographic locations.
2024	-	Multiple questions from tilt, ITCZ, wind pattern and radiation at earth surface.

4. Solar Radiation Quality

- Earth প্রায় সব energy সূর্য থেকে পায়; solar radiation climate-এর dominant influence.
- Spectrum in lecture: 290-2300 nm.
- UV: 290-380 nm photochemical effects, sunburn etc.
- Visible: 380-700 nm.
- Short infrared: 700-2300 nm radiant heat dominant.
- Shorter wavelengths-এর কিছু অংশ atmosphere absorb করে।

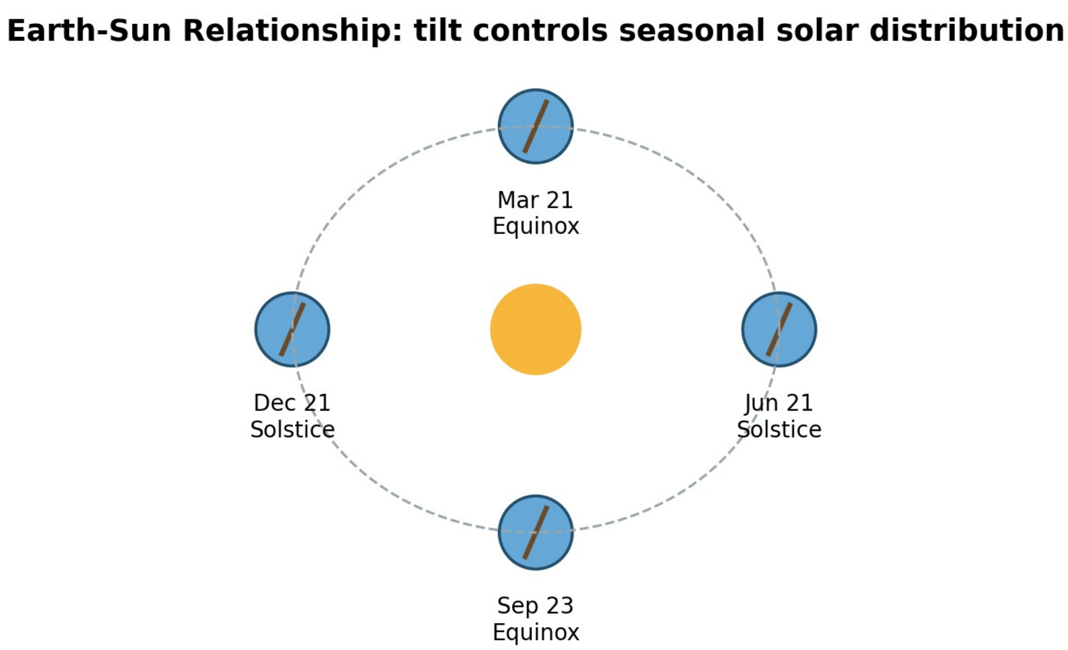
5. Solar Radiation Quantity

- Course slide solar constant = 1395 W/m^2 at upper atmosphere.
- Sun output-এর জন্য প্রায় ±2% variation; Earth-Sun distance change-এর জন্য ±3.5% variation.
- Aphelion ≈ 152 million km; perihelion ≈ 147 million km (lecture value).

Exam-safe line

Quality = wavelength / type of solar energy; Quantity = intensity / amount of solar energy.

6. Earth-Sun Relationship & Tilt of Earth's Axis



Sketch: Earth-Sun relationship, solstices and equinoxes

- Earth axis orbital plane-এর সাথে lecture slide অনুযায়ী 66.5°; equivalent tilt from the perpendicular is about 23.5°.
- Maximum solar intensity-এর zone Tropic of Cancer (around 21 June) ও Tropic of Capricorn (around 21 December)-এর মধ্যে migrate করে.
- Around 21 March and 23 September, sun’s rays equator-এ প্রায় normal equinox condition.

- Seasonal change-এর মূল exam logic: tilted axis + revolution changing solar angle + changing day length.

Long-term orbital term	Lecture idea
Eccentricity	Orbit shape changes; ~100,000-year cycle.
Obliquity	Axis angle changes; ~41,000-year cycle.
Precession	Axis “wobble”; ~26,000-year cycle.

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

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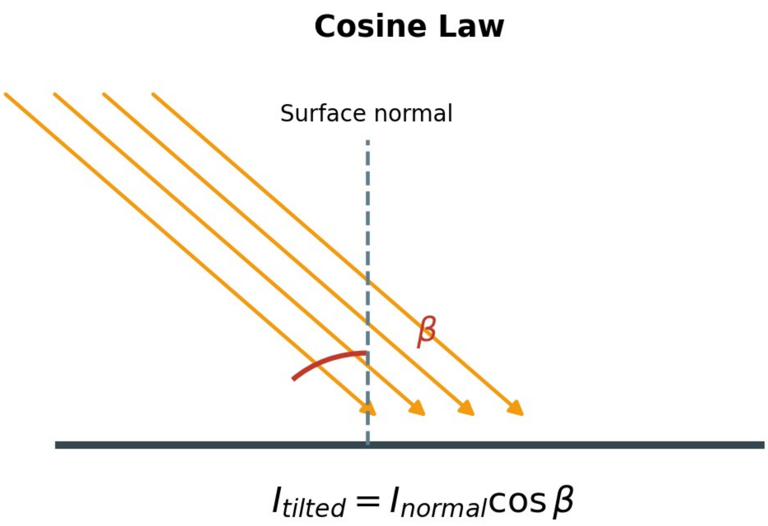
Year	Marks	Matching question / focus
2017	20	Evaluate Tilt of the Earth's Axis from viewpoint of global climate variation.
2022	12	Evaluate Earth-Sun relationship from viewpoint of global climate variations.
2024	8	Explain how tilt affects distribution of solar radiation and causes seasonal changes.

7. Radiation at the Earth's Surface

Surface-এ পৌঁছানো radiation তিনটি গুরুত্বপূর্ণ কারণে বদলায়: angle of incidence, atmospheric depletion এবং sunshine duration.

7.1 Cosine Law



If radiation is oblique, same energy spreads over a larger area; intensity falls.

$$I_{tilted} = I_{normal} \times \cos \beta$$

β = angle between solar ray and the normal to the surface

- Normal incidence energy concentrated intensity maximum.
- Oblique incidence → same radiation larger area-তে spread intensity কম।

7.2 Atmospheric Depletion

Atmosphere-এর ozone, vapour, dust ইত্যাদি radiation absorb/scatter করে; ফলে surface-এ energy কমে।

7.3 Duration of Sunshine

Daylight/sunshine কতক্ষণ পাওয়া যাচ্ছে সেটি total solar receipt-কে প্রভাবিত করে। Duration আর intensity একই জিনিস নয়—এই distinction পরে Elements of Climate-এ আবার আসবে।

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

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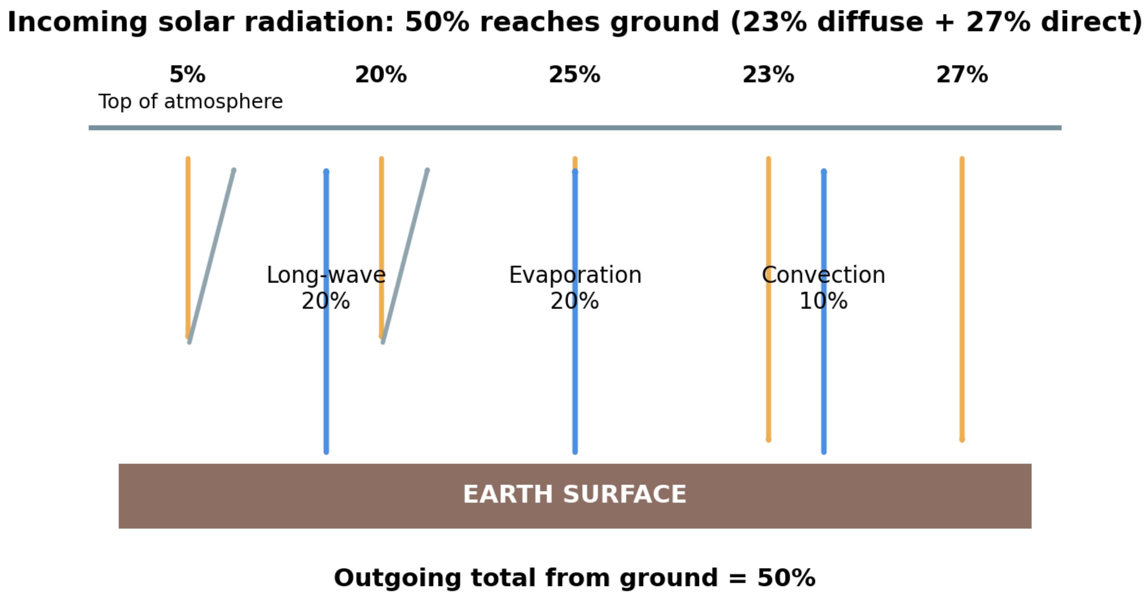
Year	Marks	Matching question / focus
2023	10	Evaluate Radiation of the Earth's Surface from global climate variation viewpoint.
2024	9	Evaluate Radiation of the Earth's Surface from viewpoint of climate variation at global scale.

8. Earth's Thermal Balance



Redrawn from lecture: incoming and outgoing percentage balance

Incoming component	Course value
Reflected from ground	5%
Reflected from cloud	20%
Absorbed in atmosphere	25%
Diffuse radiation reaching ground	23%
Direct radiation reaching ground	27%

Outgoing from ground	Course value
Long-wave radiation	20%
Evaporation, then radiation	20%
Convection, then radiation	10%
Total	50%

Core logic

Lecture balance: 50% of incoming solar energy reaches the ground (23% diffuse + 27% direct), and the ground returns 50% through long-wave radiation + evaporation + convection. This balance prevents indefinite heating.

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

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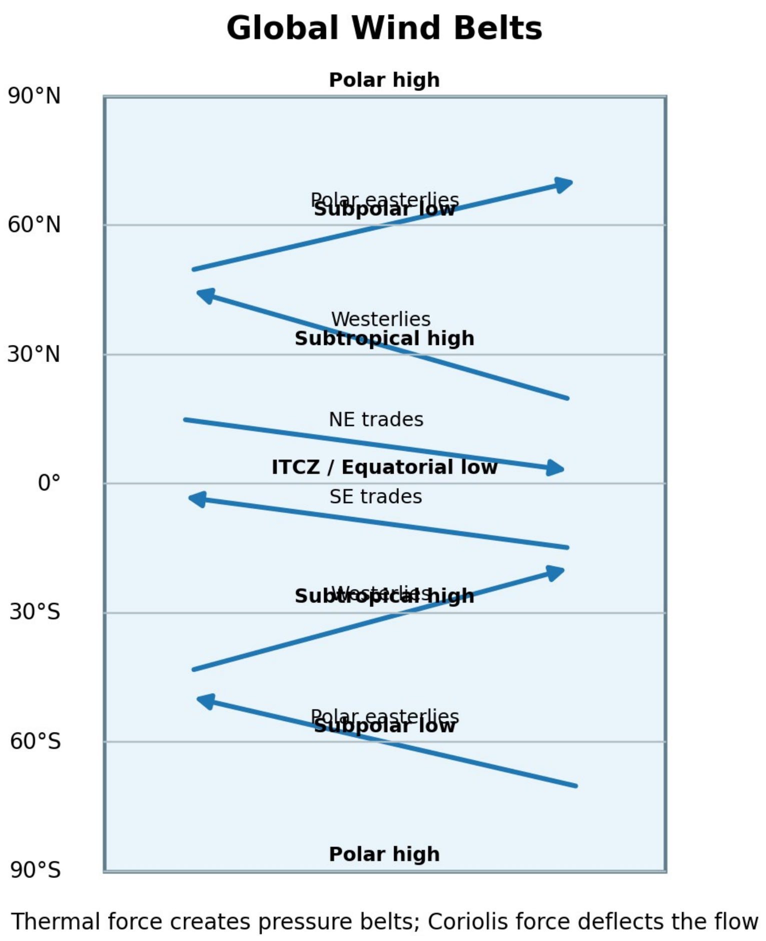
Year	Marks	Matching question / focus
2016	15	How does the earth keep thermal balance by equating incoming and outgoing radiation? Show diagram.
2021	12	How does the earth keep thermal balance by equating incoming and outgoing radiation?

9. Global Wind Pattern — mechanism



Pressure belts and prevailing surface winds

- Equator বেশি heat পায় → surface air warm, expands, becomes light rises equatorial low pressure.
- Rising warm-humid air upper level-এ subtropical zone-এর দিকে যায়; cooling/densification-এর পর ~30° latitude-এ sink করে → subtropical high.
- Subtropical high থেকে এক branch equator-এ ফেরে (trade wind), অন্য branch subpolar front-এর দিকে যায় (westerlies).
- Polar region cold/dense polar high; surface flow subpolar low-এর দিকে → polar easterlies.
- এভাবে Hadley, Ferrel, Polar circulation cells ও pressure belts তৈরি হয়।

Shortcut / মনে রাখার কৌশল — 0-30-60-90

0° = low (ITCZ), 30° = high, 60° = low, 90° = high. Pressure-belt sequence মনে রাখলে wind pattern সহজ হয়।

10. Thermal Force & Coriolis Force

Force	What it does	Exam phrase
Thermal force	Unequal heating থেকে pressure difference ও convective circulation তৈরি করে.	Motive force of wind.
Coriolis force	Earth rotation moving air-এর direction deflect করে.	Modifies wind direction.
Result	Thermal + Coriolis → trade winds / global wind pattern.	Both are needed to explain actual wind direction.

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

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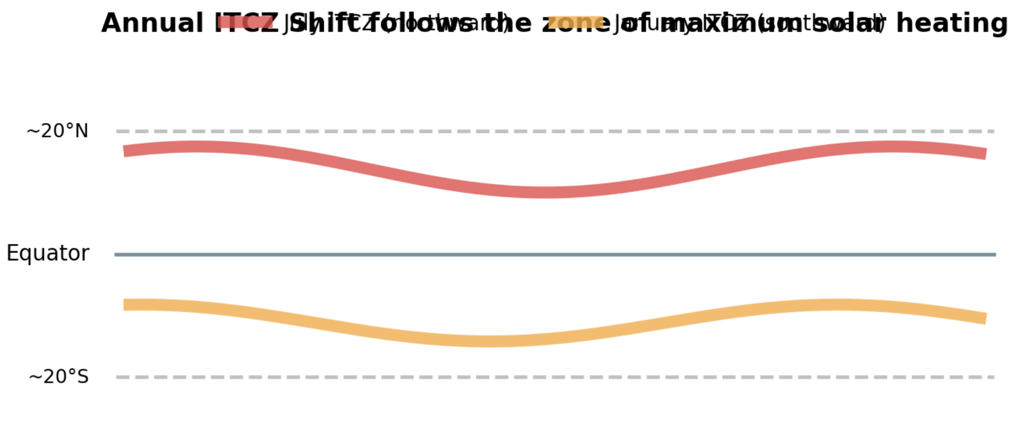
Year	Marks	Matching question / focus
2016	20	Evaluate forces interacting to generate global wind pattern at different latitudes.
2022	10	Describe relationship between thermal force and Coriolis force with diagrams.
2024	15	Explain global wind patterns at different latitudes in both hemispheres with sketches.

11. ITCZ, Doldrums & Annual Wind Shift

- At equator, northern and southern trade winds meet and air rises Inter -Tropical Convergence Zone (ITCZ).
- Surface wind may be calm/light and irregular sailors call it “Doldrums”.
- ITCZ maximum solar heating zone-কে follow করে; July-এ northward, January-এ southward shift.
- Annual shift temperature, wind direction এবং rainfall-এর seasonal change ঘটায়.



Conceptual seasonal migration of ITCZ

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

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Year	Marks	Matching question / focus
2016	15	Assess impact of annual wind shift from north to south of equator.
2021	10	Describe relationship between ITCZ and annual wind shifts with figures.
2023	10	Define doldrums and explain its location and formation.
2024	12	Why is ITCZ generally calm, and why does it shift more toward the Northern Hemisphere?

12. Mid-latitude Westerlies, Polar Winds & Topography

- Westerlies: subtropical high থেকে subpolar low-এর দিকে surface flow; Coriolis deflection gives westerly character.
- Polar easterlies: polar high থেকে subpolar low-এর দিকে cold surface air; Coriolis deflection gives easterly flow.
- Topography regional pressure/temperature/wind patternকে modify করে; land, forest, water differentially heat হয় এবং mountains wind path বদলায়.

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

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Year	Marks	Matching question / focus
2021	10	Assess influence of topography from perspective of global climate variations.
2022	13	Explain formation process of mid-latitude westerlies and polar winds.
2024	-	Global wind pattern question combines all wind belts.

LECTURE 03 — Elements of Climate & Climatic Data

1. Elements that matter to a building designer

Element	Why it matters to design
Temperature	heat gain/loss, comfort, material response
Humidity & vapour pressure	evaporation, condensation, comfort, decay
Precipitation & driving rain	roof/drainage/envelope leakage risk
Solar radiation	heat gain, daylight, shading
Wind	ventilation, heat loss, pressure/rain exposure
Sky condition	diffuse radiation/daylight
Vegetation	shade, wind filtering, surface moisture/evaporation

Shortcut / মনে রাখার কৌশল — T-H-P-S-W-S-V

Temperature, Humidity, Precipitation, Solar, Wind, Sky, Vegetation — 7 elements.

2. Climatic Information: raw data → design data

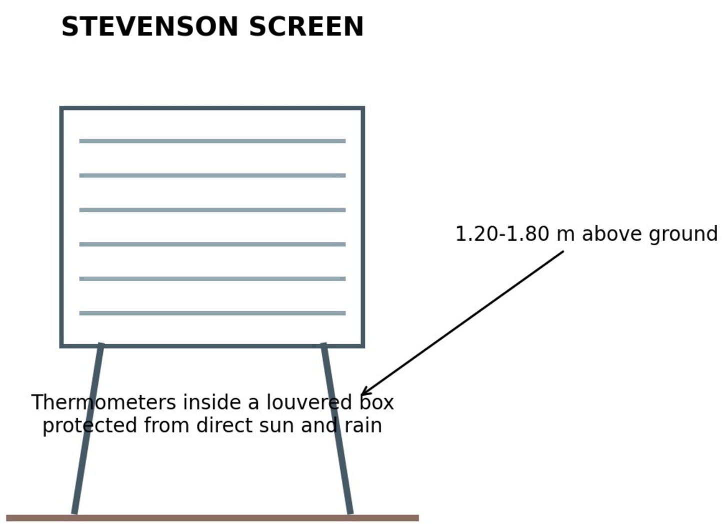
Meteorological station-এর raw data directly designer-এর answer নয়। Building designer-কে beneficial/harmful features identify করার জন্য data analyse করতে হয়।

Design logic

Record → Analyse → Identify extremes/means → Translate into design requirement.

3. Temperature Measurement

- Unit: °C; mercury thermometer commonly used in slide.
- Continuous record: thermograph.
- True air temperature shade-এ measured; thermometer louvered wooden Stevenson screen-এর ভিতরে রাখা হয়.
- Height: about 1.20-1.80 m above ground (lecture slide).
- Useful data: daily max/min + diurnal range; monthly mean/extremes; annual mean/extremes/range.



Stevenson Screen — clean exam sketch

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

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Year	Marks	Matching question / focus
2021	5	Short note: Stevenson Screen.
2022	10	Which instrument is required to measure temperature in open field? Show working mechanism of Stevenson Screen.

4. Humidity — AH, RH & Vapour Pressure

Term	Definition / formula	Unit
Absolute Humidity (AH)	Actual moisture present per unit mass/volume of air.	g/kg or g/m³
Relative Humidity (RH)	Actual moisture / moisture-holding capacity at same temperature × 100	%
Vapour pressure (Pv)	Partial pressure due to water vapour in air.	N/m²

$$RH = AH / SH \times 100\ (\%)$$

SH = saturation humidity / moisture-holding capacity at that temperature

$$P = Pa + Pv$$

Atmospheric pressure = partial pressure of dry air + partial vapour pressure

- RH measured with wet-and-dry bulb hygrometer; readings can be interpreted using psychrometric chart.
- Humidity data can be analysed as daily, monthly and annual maxima/minima/means.

Bangla intuition

AH বলে “আসলে কত জলীয়বাষ্প আছে”; RH বলে “এ তাপমাত্রায় যতটা ধরতে পারত, তার কত শতাংশ ভরা আছে।”

5. Precipitation

- Rain, hail, dew, snow, frost—atmosphere থেকে deposited water-এর collective term.
- Measured by rain gauge; unit commonly mm/day or mm/month.
- Monthly pattern dry/wet season দেখায়.
- Maximum 24-hour rainfall flooding risk-এর guide; maximum hourly rainfall intensity surface drainage design-এর জন্য প্রয়োজন.

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

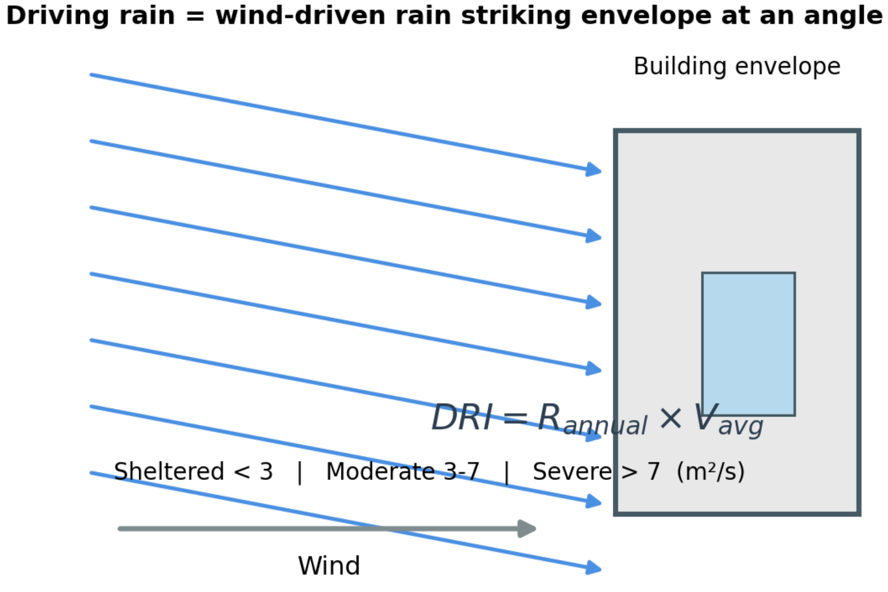
নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2024	10	Describe the process of measuring climatic parameters—temperature, humidity and air flow—from a specific site for design climatic response.
2024	8	What climatic factors affect building design and how do they influence it?

LECTURE 03.1 — Elements for Building Designers

1. Driving Rain & Driving Rain Index (DRI)

Driving rain = wind-driven rain যা verticalভাবে না পড়ে angle-এ building envelope-এ আঘাত করে এবং joints, window reveals, cracks/gaps দিয়ে water penetration ঘটাতে পারে।



Driving rain mechanism and DRI thresholds

DRI = Annual Rainfall (m) × Annual Average Wind Speed (m/s)

Unit in the lecture: m^2/s

Exposure	DRI	Design meaning
Sheltered	3 m^2 / s	Low combined wind-rain exposure; normal detailing
Moderate	3-7 m^2/s	More careful joints, sills, reveals and weatherproofing
Severe	> 7 m^2/s	Highly exposed/coastal/hilltop; robust envelope and gap control

Worked example from slide

Annual rainfall 1.2 m × average wind 4.0 m/s = DRI 4.8 m^2/s → Moderate exposure.

Shortcut / মনে রাখার কৌশল — R × V

DRI মনে রাখো: Rainfall × Velocity. বেশি rain বা বেশি wind—দুটিই exposure বাড়ায়।

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2021	5	Short note: Driving Rain.
2022	10	Define driving rain. Explain how to design exposure of a building using DRI.

Year	Marks	Matching question / focus
2023	10	Define driving rain. Explain building exposure design using driving rain index.
2024	7	Define driving rain. Write in detail how to design exposure of a building using DRI.

2. Sky Condition & Cloud Cover

- Sky condition mainly cloud presence/absence দিয়ে described হয়.
- Typical observation around 09:00 and 15:00; night records comparatively limited.
- Cloud cover tenths/eighths (octas) বা percentage-এ expressed হতে পারে: 50% = 5/10 = 4/8.
- Designer-এর observation time/frequency জানা দরকার, কারণ sky condition diffuse radiation ও daylight influence করে.

3. Solar Radiation — Sunshine Duration vs Radiation Intensity

Item	Sunshine Duration	Radiation Intensity
Meaning	How long bright sun shines	Actual radiant power per unit area
Typical unit	hours/day, monthly average	W/m^2; also $Btu/ft^2 \cdot h$, $kcal/m^2 \cdot h$
Instrument	Sunshine recorder (Campbell-Stokes type)	Pyranometer / pyrheliometer / actinometer / solarimeter
Design use	Long-term sunshine availability	Directly relates to solar heat gain

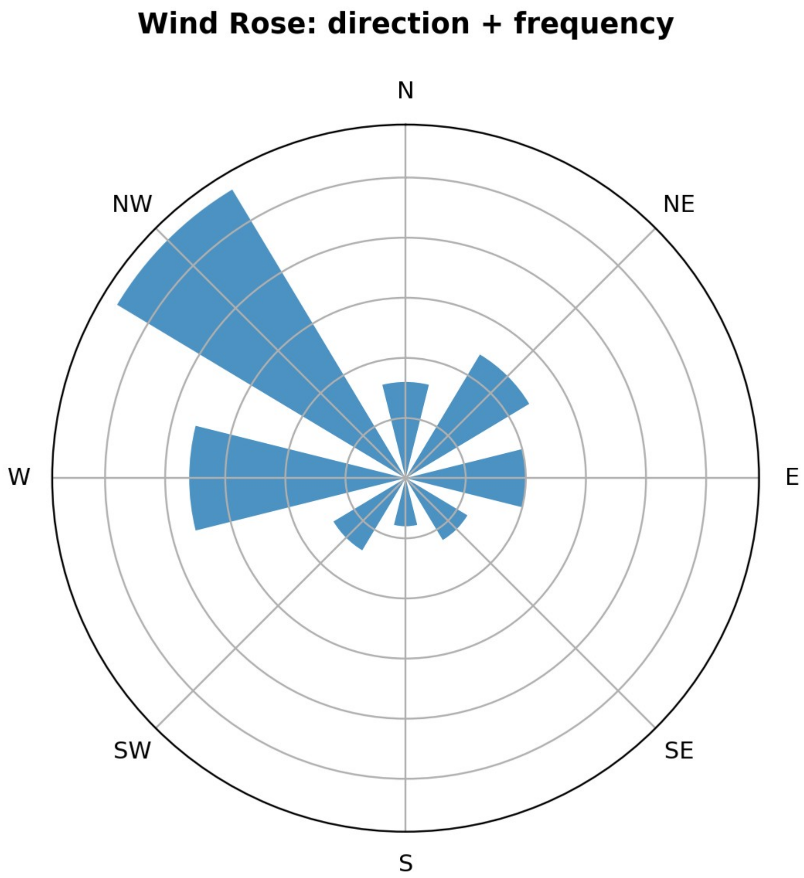
Very important distinction

Zero bright-sunshine hours does NOT mean zero solar energy: an overcast sky can still deliver diffuse radiation.

4. Wind Measurement

- Wind velocity: cup-type / propeller anemometer.
- Wind direction: wind vane.
- Continuous velocity + direction record: anemograph.
- Free wind: open flat country at ~10 m height; urban reading ~10-20 m to reduce obstruction effects.
- Unit: m/s; other units may include ft/min, mph, knot.

5. Wind Rose — how to read



Longer spoke = wind occurs more frequently from that direction

- Direction: spoke যে দিক দেখায়, wind সেই direction থেকে এসেছে.
- Frequency: spoke length / radial scale দ্বারা বোঝা যায়.
- Speed classes: colour/segment দ্বারা বিভিন্ন speed range দেখানো যায়.
- “Calm” percentage center-এ separately দেওয়া হতে পারে.

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2017	20	Draw monthly and annual wind-frequency graphs and explain their interpretation.
2024	13	Draw monthly and annual wind frequency graph and explain how they are usually interpreted.

Source-note

The Section B “Elements for Building Designers” slide lists vegetation as an element but does not provide a separate detailed vegetation lecture. Vegetation effects are discussed inside tropical-climate profiles and site-climate factors later in this note.

LECTURE 04 — Classification of Tropical Climate

1. Basis of classification

- Lecture credits G. A. Atkinson (1953).
- Basis: extremes of temperature and humidity affecting human comfort.
- Tropical climates are divided into three major zones plus three sub-groups (six named types in the slide).

Major / subgroup types
Warm-humid equatorial climate
Warm-humid island or trade-wind climate
Hot-dry desert or semi-desert climate
Hot-dry maritime desert climate
Composite or monsoon climate
Tropical upland climate

Shortcut / মনে রাখার কৌশল — 3 pairs

Warm-humid (equatorial + island) | Hot-dry (desert + maritime) | Seasonal (composite + upland).

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2021	10	Explain salient features affecting classification of tropical climates as suggested by G. A. Atkinson.
2023	10	Write classification of tropical climate and explain how climate zones are formed on earth.

2. Warm-Humid Equatorial Climate

Parameter	Lecture profile
Location	roughly within ~15° N and S
Air temperature	Mean max 27-32°C; mean min 21-27°C; narrow diurnal/annual range
RH	~75%, may vary ~55-100%
Vapour pressure	~2500-3000 N/m²
Rainfall	~2000-5000 mm/year; severe short storm may ~100 mm/h
Sky	Cloudy ~60-90%; glare possible
Solar radiation	Partly reflected/scattered by clouds; diffuse component important
Wind	Generally low, but strong squalls possible
Vegetation / ground	Rapid growth, high water table/waterlogging possible
Special risks	Mould, algae, rust, rot, insects

Shortcut / মনে রাখার কৌশল — গরম-ভেজা-মেঘ-বৃষ্টি

Warm-humid = high humidity + heavy rain + cloudy sky + small temperature range. “দিন-রাত খুব বেশি বদলায় না।”

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2016	20	Compare characteristics of Hot-dry and

Year	Marks	Matching question / focus
		Warm-humid climate.
2023	10	Describe features of warm-humid equatorial climate with examples.
2024	16	Why is climate warm and humid between equator and about 15° N/S? Explain using climatic features.

3. Hot-Dry Desert / Semi-Desert Climate

Parameter	Lecture profile
Location	approximately 15-30° N and S
Hot-season day max	43-49°C
Hot-season night min	24-32°C
Cold-season night min	10-18°C
Diurnal range	Very high: ~17-22°C
RH	Low: ~10-55%
Rainfall	Very low/variable: ~50-155 mm/year; long dry periods possible
Sky	Mostly clear; dust haze may create glare
Solar radiation	Strong/direct by day; strong night re-radiation under clear sky
Wind	Hot, local, dusty; dust storms possible
Vegetation	Sparse; dry soil; low water table
Material issue	Rapid day-night temperature swing may crack materials

Shortcut / মনে রাখার কৌশল — দিন আগুন—রাত ঠান্ডা

Hot-dry-এর exam identity: very hot day + cooler night + low RH + clear sky + strong direct sun + little rain + dust.

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2016	20	Compare Hot-dry with Warm-humid climate.
2022	15	Describe features of composite/monsoon and hot-dry desert climatic patterns with examples.
2024	12	How does hot desert form in subtropical region and why is day-night temperature range large?

4. Composite / Monsoon Climate

- Large land masses near tropics; marked seasonal changes in solar radiation and wind direction.
- Usually two dominant seasons; lecture says roughly two-thirds hot-dry and one-third warm-humid; sometimes cool-dry third season.
- Examples on slide: Lahore, Mandalay, Asuncion, Kano, New Delhi.

Season	Day mean max °C	Night mean min °C	Diurnal range °C
Hot-dry	32-43	21-27	11-22
Warm-humid	27-32	24-27	3-6
Cool-dry	Up to 27	4-10	11-22

- Dry period RH ~20-55%; wet season RH ~55-95%.
- Monsoon rain intense/prolonged; annual rainfall lecture range ~500-1300 mm.
- Sky: overcast/dull in monsoon; clear/dark blue in dry season.
- Wind: dry season hot/dusty; monsoon wind fairly strong and steady.
- Seasonal humidity change may weaken materials; dust/sand storms may occur.

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2017	20	Characterize composite or monsoon climate with example.
2021	10	Characterize composite or monsoon climate with example.
2022	15	Features of composite/monsoon and hot-dry desert climate.
2023	10	Passive control in tropical climate is more critical than subtropical/polar climate — justify.

5. Warm-Humid vs Hot-Dry vs Composite — one-page revision table

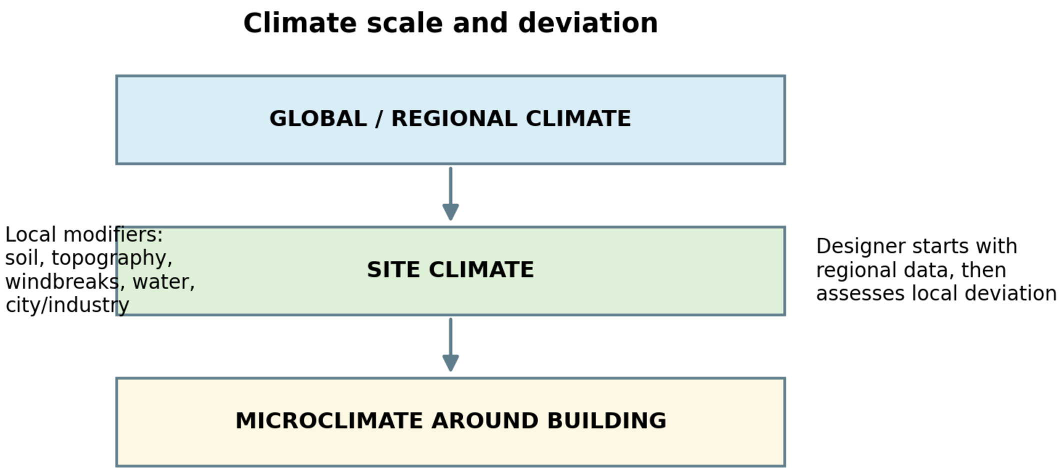
Feature	Warm-humid	Hot-dry	Composite/monsoon
Temperature range	Narrow	Large diurnal	Season-dependent
Humidity	High	Low	Low dry / high wet
Rain	High	Very low	Strong monsoon season
Sky	Cloudy	Mostly clear / dust haze	Seasonal
Solar	Diffuse + cloud-modified	Strong direct	Strong in clear season
Wind	Low except squalls	Hot/dusty/local	Strong monsoon + dusty dry
Main memory	Heat + moisture	Heat + dryness	Seasonal combination

Source-note

The Section B tropical-climate slides focus on climatic characteristics. Detailed building-design prescriptions are not independently developed in this slide set; where older questions ask protective measures, revise the relevant design-control material from the course/reference material separately.

LECTURE 05 — Regional, Site & Urban Climate

1. Macroclimate, Site Climate & Microclimate



Regional data is the starting point; local modifiers create site-specific deviation.

Scale	Meaning
Regional / macroclimate	Broad climate described by regional/met station data.
Site climate	Climate of the actual area available for the project, horizontally and vertically.
Microclimate	Very local climate around a building/space; scale depends on discipline.

Important

Macroclimate is useful as a guide, but lecture says it is seldom sufficient in accuracy for a particular site.

2. How to assess site climate for a large site

6. Start with regional climatic data.
7. Identify dominant local factors and expected deviations.
8. Use available site observation/measurement where possible.
9. Take advice from experienced local observer/expert if long observation is impossible.
10. Select the area most suitable for habitation / project purpose.

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2020	10	Describe fundamental process of designing a site as a designer using site climate knowledge in a large site.
2022	10	Describe fundamental process of designing a site using site climate knowledge.
2023	10	Describe fundamental process of designing a residential building at KUET using site climate knowledge.

3. Factors Constituting Regional Climate

Level	Factors
Primary	Global climatic factors; detailed slide highlights temperature, humidity, air flow as primary climatic outcomes.
Secondary	Location of region; diurnal variation; altitude.
Tertiary	Soil condition; topography; natural wind breakers; man-made wind breakers; water bodies; cities/industries; deserts.
Local & minor	Ocean current; sky condition; smog; smoke; fog; thunder/lightning; sand storms.

Shortcut / মনে রাখার কৌশল — P-S-T-L

Primary → Secondary → Tertiary → Local. প্রথমে hierarchy লিখলে বড় প্রশ্নের structure নিজে থেকেই তৈরি হয়।

4. Key local modifiers — quick explanations

Factor	Effect
Soil condition	Stony/sandy surface quickly heats and reradiates; fine/organic soil behaves differently.
Topography	Cool heavy air settles in depressions → locally cooler/humid conditions.
Natural wind breaker	Mountains/large trees obstruct or redirect wind.
Ocean current	Warm/cool currents transport heat and modify coastal temperature/humidity/rainfall.
Smog / smoke / fog	Modify humidity, radiation, visibility and air quality.
Sand storm	Openings/ventilators require protection; façade treatment may respond to dust/sand.

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2016	25	Identify local factors responsible for developing site/microclimate; distinguish from regional/macro climate.
2017	10	Short notes: Ocean current / Natural wind breaker.
2020	20	Explain factors that contribute to formation of site climate.
2024	10	What is site climate, and what factors cause deviations in local climate?

5. Deviation Factors

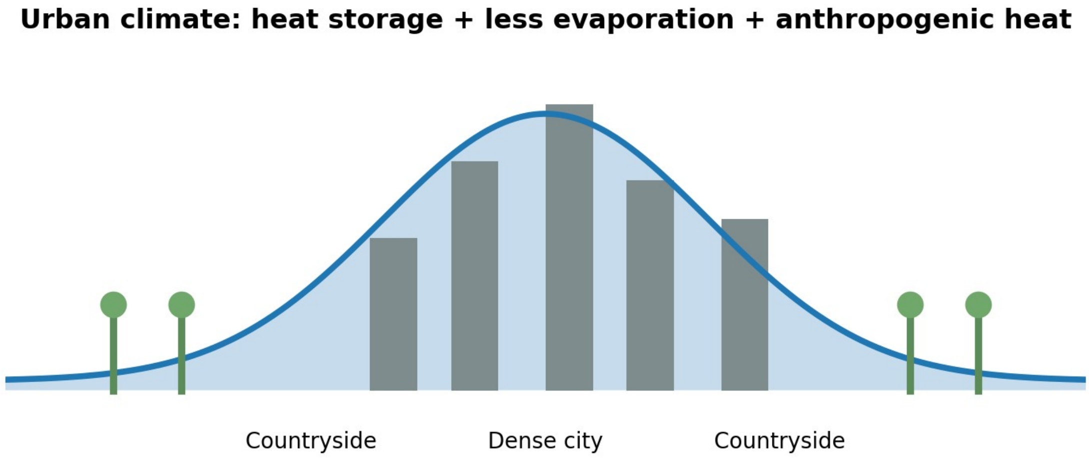
- Air temperature
- Temperature inversion
- Humidity
- Precipitation
- Sky condition
- Solar radiation
- Air movement / wind velocity gradients
- Special characteristics
- Vegetation

Source-note

The uploaded site-climate slide lists “temperature inversion” as a deviation factor but does not separately define or derive it. Therefore this note does not invent a detailed mechanism beyond the source. In an exam, identify it as a local/vertical temperature-structure deviation factor unless your teacher has given a fuller class explanation.

6. Urban Climate

- Changed surface qualities: pavements/buildings absorb more solar energy; evaporation decreases.
- Buildings cast shadows, block/channel wind, store heat and release it slowly.
- Energy seepage: AC/refrigeration, electrical appliances, engines and industry add heat.
- Atmospheric pollution forms a barrier to outgoing radiation.



Conceptual urban heat-island profile

Effect reported in slide	Value / behaviour
City-countryside temperature deviation	up to about 8°C
Relative humidity in urban area	about 5-10% lower
Wind velocity	may drop to less than half of adjoining countryside
Flow around buildings	turbulence / eddies at leeward corners

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2021	10	Distinguish between urban and macroclimate.
2022	15	Analyze factors responsible for deviation of urban climate and effects of deviation.
2023	15	Analyze factors responsible for deviation of urban climate and its effects.
2024	8	Urban climate depends on human intervention — explain.

LECTURE 06 — Thermal Comfort & Human Heat Balance

1. Why Thermal Comfort matters

- Daily life cycle = activity fatigue recovery ; unfavorable climate এই cycle disturb করতে পারে.
- Thermal stress discomfort, reduced efficiency এবং health breakdown ঘটাতে পারে.
- Therefore building designer must understand both climate and how the human body acts/reacts to heat.

Definition idea

Thermal comfort = এমন thermal condition যেখানে মানুষের body heat balance বজায় থাকে এবং ব্যক্তি পরিবেশকে thermally acceptable মনে করে. Lecture emphasis is on physiological balance and comfort variables.

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2023	10	Describe significance of thermal comfort for human body.
2024	12	What is thermal comfort? How does the human body maintain thermal balance?

2. Metabolism & Heat Production

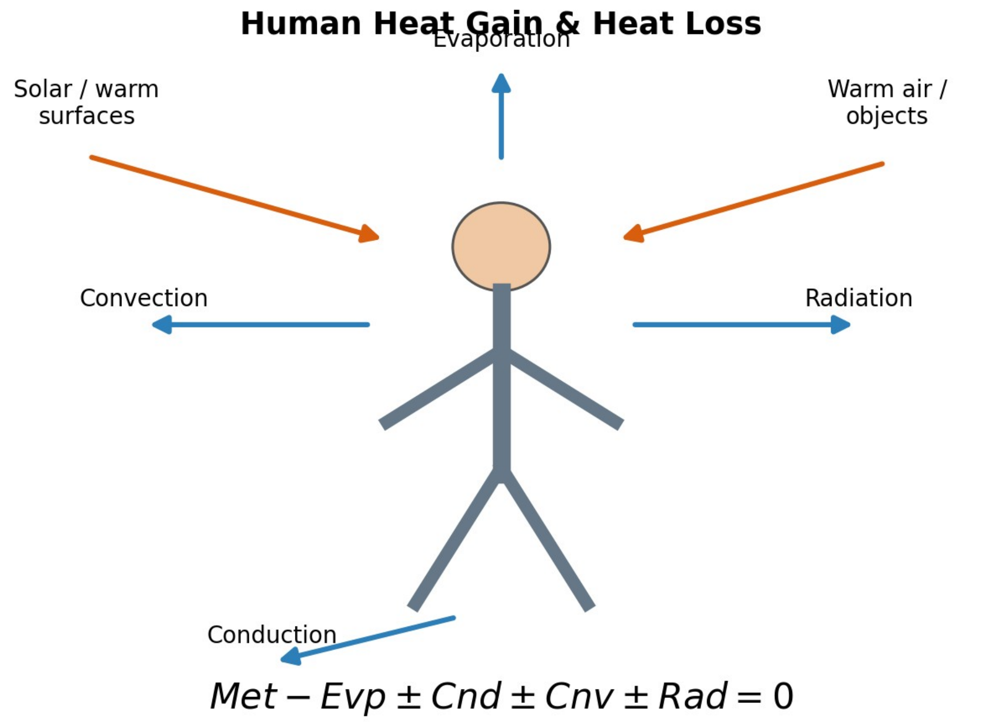
- Metabolism = foodকে living matter ও energy-তে convert করার process.
- Basal metabolism = continuous automatic/vegetative process.
- Muscular metabolism = consciously controlled muscular work.
- Lecture slide states roughly 20% produced heat is used inside body and remaining 80% surplus heat must be dissipated.

Activity	Heat production (W) — lecture table
Sleeping	Min. 70
Sitting / typing / light bench work	130-160
Sitting, heavy arm/leg movements	190-230
Standing moderate work / some walking	220-290
Walking + moderate lifting/pushing	290-410
Intermittent heavy lifting/digging	440-580
Hardest sustained work	580-700
Maximum heavy work ~30 min	Max. 1100

Source-note

Billah Sir class note marks the heat-production table as “not need to memorize”; understand the trend: more activity → higher metabolic heat.

3. Human Heat Gain & Heat Loss



Redrawn heat-exchange sketch: conduction, convection, radiation, evaporation

Mode	Heat gain / loss logic
Metabolism	Internal heat production.
Conduction	Direct contact; depends on surface temperature difference.
Convection	Heat exchange with air touching skin/clothing; faster air movement usually increases convective exchange.
Radiation	Heat exchange with surrounding surfaces / sun without direct contact.
Evaporation	Sweat + imperceptible perspiration + breathing; dry air allows faster evaporation.

Shortcut / মনে রাখার কৌশল — C-C-R-E

Heat-loss routes: Conduction, Convection, Radiation, Evaporation. Exam sketch-এ চারদিকে arrows দিলে marks বাড়ে।

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2018	20	What is thermal comfort? How does the human body produce and release heat to the environment?
2020	15	Describe human body heat exchange with environment through conduction, convection, radiation and evaporation.
2022	10	Describe human body mechanism of heat production and subjective variables of thermal comfort.

4. Body temperature & steady state

- Deep body temperature must remain balanced around 37°C.

- Skin/surface temperature is about 2°C lower in the lecture slide.
- All surplus heat must be dissipated to maintain steady condition.

5. Human Thermal Balance Equation

$$\text{Met - Evp} \pm \text{Cnd} \pm \text{Cnv} \pm \text{Rad} = 0$$

Metabolism - evaporation ± conduction ± convection ± radiation = thermal steady state

Sign idea

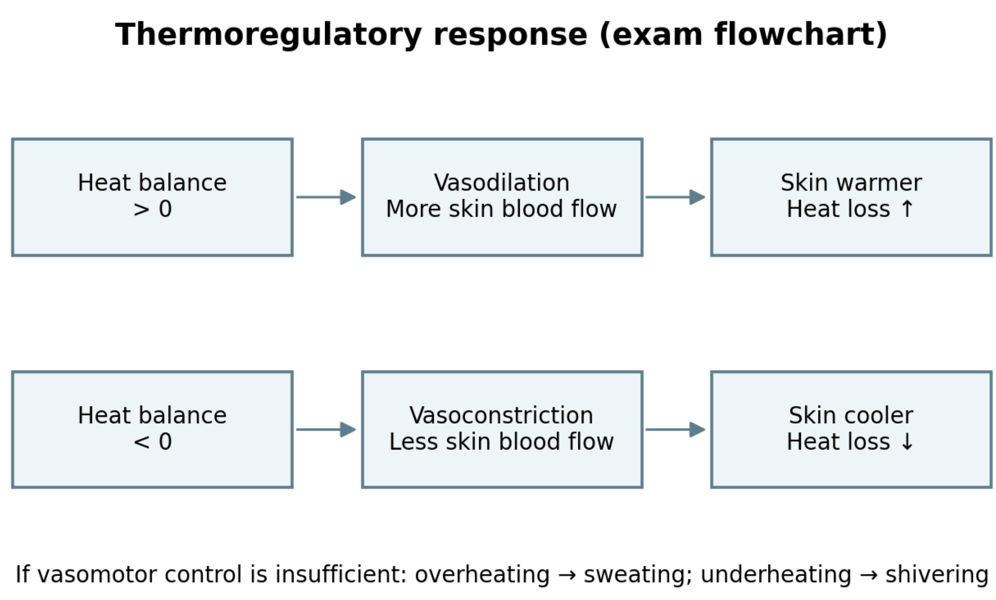
Conduction / convection / radiation can be gain or loss depending on direction of heat flow, তাই ± sign. Evaporation is written as heat loss in the lecture equation.

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2021	13	Write down the thermal balance equation of human body. What happens if the equation is unbalanced?
2023	12	Write thermal balance equation and describe regulatory mechanisms when it is unbalanced.

6. Vasomotor Adjustment, Sweating & Shivering



Thermoregulatory response flowchart

- If heat balance > 0: skin blood circulation increases skin temperature rises heat -loss processes accelerate (vasodilation-type response).
- If heat balance < 0: skin circulation reduces skin temperature falls heat -loss processes slow (vasoconstriction-type response).
- If overheating continues despite vasomotor regulation sweating starts.
- If underheating continues violent shivering can sharply increase metabolic heat production.

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2023	8	Briefly explain process of vasomotor adjustment and autonomous nervous-system control of internal temperature.
2024	10	Explain concept of vasomotor adjustment as part of human thermal regulation.

7. Objective Variables of Thermal Comfort

Objective / climatic variable	What changes
Radiation	Radiant exchange with surrounding surfaces / sun.
Air temperature	Convection direction and magnitude.
Humidity	Evaporative cooling potential.
Air movement	Convection + evaporation rate.

Shortcut / মনে রাখার কৌশল — R-A-H-A

Radiation – Air temperature – Humidity – Air movement. “RAHA” বলে মনে রাখো।

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2023	8	Describe objective variables by which thermal comfort is affected.
2024	-	Thermal comfort question expects variables + heat balance mechanism.

8. Subjective / Individual Variables

Variable	Bangla meaning / idea
Clothing	পোশাকের insulation
Acclimatization	দীর্ঘদিনের climate adaptation
Age & sex	metabolic/physiological differences
Body shape	surface-area relation
Subcutaneous fat	ত্বকের নিচের insulation
State of health	physiological condition
Food & drink	metabolism / hydration effect
Skin color	listed in lecture as an individual variable

Shortcut / মনে রাখার কৌশল — কাপড়-অভ্যাস-বয়স-গঠন-চর্বি-স্বাস্থ্য-খাবার-ত্বক

Subjective variables sequence বাংলা শব্দে মুখস্থ করো; English term পাশে লিখলে examiner-friendly হয়।

Question Bank Match — পরীক্ষায় যেভাবে এসেছে

নিচের wording পাঠযোগ্য করার জন্য সামান্য normalised করা হয়েছে; বছর ও marks স্ক্যানে যতটা স্পষ্ট, ততটাই রাখা হয়েছে।

Year	Marks	Matching question / focus
2018	5	Outline various subjective variables by which thermal preferences are influenced.
2022	10	Human heat production + subjective variables.
2024	5	Outline various subjective variables influencing thermal preferences.

FINAL RAPID REVISION — Section B

1. Formula Sheet

Formula	Use
$I_{tilted} = I_{normal} \cos \beta$	Cosine law / oblique solar incidence
DRI = Annual rainfall × Average wind speed	Driving-rain exposure
RH = AH / SH × 100%	Relative humidity
$P = P_a + P_v$	Atmospheric pressure components
Met - Evp ± Cnd ± Cnv ± Rad = 0	Human thermal balance

2. Must-draw Sketch List

Priority	Sketch
★★★★★	Global wind belts + pressure zones / ITCZ
★★★★★	Earth thermal balance incoming/outgoing
★★★★★	Human heat gain/loss + heat balance
★★★★☆	Earth tilt / solstice-equinox
★★★★☆	Driving rain + DRI exposure
★★★★☆	Stevenson screen
★★★★☆	Wind rose interpretation
★★★☆☆	Urban heat-island / site climate hierarchy
★★★☆☆	Cosine law

3. Highest-repeat Answer Blocks

Block	Prepare these together
A — Global circulation	Thermal force + Coriolis + ITCZ + trade winds + westerlies + polar easterlies + annual shift
B — Tropical climate	Classification + warm-humid + hot-dry + composite + comparison
C — Site climate	Regional vs site + factors hierarchy + urban climate + deviation effects
D — Thermal comfort	Metabolism + 4 heat modes + heat-balance equation + vasomotor + objective/subjective variables
E — Climate measurement	Stevenson + humidity terms + precipitation + DRI + solar instruments + wind rose

4. Common Exam Traps

- Climate এবং weather এক করা যাবে না.
- Sunshine duration আর radiation intensity এক জিনিস নয়.
- ITCZ “এক জায়গায় fixed” নয়—seasonally shifts.
- DRI-তে rainfall ও wind speed দুটোই লাগে.
- Warm-humid-এ diurnal temperature range narrow; hot-dry-এ large.
- Site climate = regional climate copy নয়; local deviation analyse করতে হয়.
- Thermal balance equation-এ Cnd/Cnv/Rad gain বা loss হতে পারে, তাই ±.

Last-minute rule

একটি 10-mark question-এর জন্য minimum: definition + 5-6 clear points + one sketch/table + 1 design significance line. 15-20 marks হলে mechanism, comparison, numerical values এবং diagram detail বাড়াও।